



Figure 1: CuO hierarchical surfaces were fabricated as follows: (1) Ti/Cu/Ti layers were deposited on a silicon substrate. (2) A thick ($\approx 100\ \mu\text{m}$) PR mold was patterned using photolithography. (3) The top Ti layer was removed to expose the Cu layer and then Cu micropillars were electroplated. (4) After removing the photoresist, Cu micropillars were oxidized in an alkali solution composed of NaClO_2 , NaOH , $\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$ to form CuO nanostructures.

Surface Characterization